MILITARY SPECIFICATION
REMOVER, AIRCRAFT CANOPY, P/N 2218-18
OVERHAUL OF

1. SCOPE

1.1 This specification covers the overhaul requirements for Remover, Aircraft Canopy, P/N 2218-18.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-G-4343	- Grease, Pneumatic System
MIL-S-5002	- Surface Treatment, and Metallic Coatings for Metal Surfaces of Weapons Systems
MIL-P-8585	- Primer, Coating, Zinc Chromate, Low - Moisture Sensitivity
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw
MIL-L-8937	- Lubricant, Solid Film, Heat Cured
MIL-C-26047	- Coating, Nickel - Phosphorus, Electroless, Nickel Requirements for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Card

FSC 1377

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2.1 (Cont'd)

DRAWINGS

Munitions Command

C9233331	- Carton, Packing Ammunition for Remover, Aircraft Canopy, P/N 2218-18
C9233334	- Box, Packing Ammunition for Remover, Aircraft Canopy, P/N 2218-18

Talley Industries (Code Ident. No. 98708)

2218-18	- Actuator, Canopy, Electro, Mechanical Ballistic Emergency Removal

PUBLICATIONS

Air Force Technical

T.O. 11P4-11-3	- Overhaul Instructions with Parts Breakdown, Canopy Actuator, P/N 2218-18

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other Publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA B46.1	- Surface Texture,

(Application for copies should be addressed to the Amercian Standards Association, 10 East 40th Street, New York, N.Y. 10016)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified in T.O. 11P4-11-3, and referenced specifications and standards.

3.2 Assembly torque.- The assembly torque for the installation of remover component parts shall be in accordance with requirements specified in Figure 2-7 of T.O. 11P4-11-3.

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3.3 Remover assembly.- The overhauled remover shall contain new parts in accordance with Table I:

TABLE I

Name of Part	Part Number
Seal, O-Ring	MS9021-010
Seal, O-Ring	MS9021-012
Seal, O-Ring	MS9021-021
Seal, O-Ring	MS9021-029
Seal, O-Ring	MS9021-030
Seal, O-Ring (2 ea req'd)	MS9021-034
Seal, O-Ring	MS9021-135
Seal, O-Ring	2-25 (Tetraflour, Inc.)
Seal, O-Ring	2-26 (Tetraflour, Inc.)
Seal, O-Ring	2-29 (Tetraflour, Inc.)
Seal, O-Ring	2-30 (Tetraflour, Inc.)
Seal, O-Ring	2-31 (Tetraflour, Inc.)
Basket, Cover	2218-53 (Talley Ind.)
Gasket	2218-37 (Talley Ind.)
Grommet	MS35490-4
Spring	MS28585-219
Spring, Compression	1145-57 (Talley Ind.)
Cartridge	1146-221 (Talley Ind.)
Bushing (2 ea req'd)	2218-67 (Talley Ind.)
Retainer	2218-24 (Talley Ind.)
Pin, Retaining	2218-111 (Talley Ind.)
Shear, Pin	1605-50 (Talley Ind.)
Pin, Straight	2218-60 (Talley Ind.)
Wiper, Tube	2218-25 (Talley Ind.)
Wiper, Tube	2218-27 (Talley Ind.)
Label, Identification	10536008
Plate, Instruction	1424-128 (Talley Ind.)
Label, Warning	8593294
Pin, Cotter	AN281-15-12 (replace w/latest Std. MS24665-88
Set Screw	AN565D1032H5
Set Screw	AN565A4H2
Plug, Shipping	PD40 (Protective Closures, Inc.)
Cap, Plug	12 (Protective Closures, Inc.)

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3.4 Functional

3.4.1 Motor.- The motor shall comply with the requirements of paragraph 3.3 of T.O. 11P4-11-3.

3.4.2 Operating load.- The remover shall be cycled electrically in accordance with the alternate Actuator Test Procedure, paragraph 3-2, T.O. 11P4-11-3. The operating loads shall comply with the load traces as depicted in Figure 3-2 of T.O. 11P4-11-3. The time required for linear extension, linear retraction and motor current drain shall comply with requirements in paragraph 3-2, T.O. 11P4-11-3. The requirements (paragraphs 3-2 f and 3-2 g, T.O. 11P4-11-3) for cycling the remover under opposing retraction loads shall not apply. Instead, the retraction load requirements specified in 3.4.2.1 below shall apply.

3.4.2.1 The remover shall be capable of being fully retracted while the item is being subjected to the opposing loads (Condition III, maximum tension load, Figure 3-2 of T.O. 11P4-11-3). At 28VDC input the time to fully retract the item shall not exceed 23.5 seconds and the motor current shall not exceed 10 amperes, while the item is being subjected to the opposing loads.

3.4.3 Bearing center dimensions.- The bearing center dimensions of the remover while subjected to the normal operating load shall comply with requirements of paragraph 3-1 f (1), (2), (3) and (4) of T.O. 11P4-11-3.

3.4.4 Air firing.- The remover shall stroke in accordance with the requirement of paragraph 3-1 j, T.O. 11P4-11-3 when 375 to 450 psi max air pressure is applied to the firing head.

3.5 Protective finish.- Protective finish shall be in accordance with T.O. 11P4-11-3, applicable specifications and drawings, and shall apply to all surfaces of the component, regardless of configuration. Exterior components shall have the existing protective finish stripped and the specific protective finish reapplied (See 6.1).

3.6 Workmanship.- The requirements for workmanship are as specified in T.O. 11P4-11-3, referenced specifications and the following:

3.6.1 Metal defects.- The part shall be free from cracks, splits, or other metal defects.

3.6.2 Burr.- The part shall not have burrs which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

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3.6.3 Foreign matter.- The part or assembly shall not contain chips, dirt, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550 except as specified herein.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.2.1.2 Size of lot.- A lot shall consist of an identifiable group of assembled units containing metal parts assemblies provided the homogeneity of the rebuild process has been retained. The assembly shall be loaded with accepted cartridges.

4.2.2 Examination.- One hundred percent examination shall be performed for all critical defects and defects indicated in paragraph 2.4, a thru i, T.O. 11P4-11-3 unless otherwise specified. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans and criteria in Tables II and III. For an individual defect (single characteristic), the criteria in Table II shall be used. For total defects within a major or minor group, the criteria in Table III shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

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4.2.2.1 Reclaimed parts.- Examination of reusable parts, which require no reprocessing shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be performed for all such defects. All nonconforming parts shall be rejected.

4.2.2.2 Reprocessed parts.- Examination shall be limited to a visual type inspection unless otherwise specified. All nonconforming material shall be rejected.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

TABLE III

Total Allowable Defects Per Sample Size

Sample Sizes for Major Defects	110	150	225	300
Number of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over - 43	6	8	11	14
Sample Sizes for Minor Defects	**110**	**150**	**225**	**300**
Number of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over - 48	8	11	17	20

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4.2.2.3 Clevis, Mounting, Drawing 2218-22 covering a detail of Drawing 2218-18

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Pitch diameter of thread, max.	Gage
102.	Minor diameter of thread, max.	Gage
103.	Diameter of inline holes	Gage
104.	Metal defective	Visual
105.	Protective finish inadequate or defective	Visual
106.	Surface finish improper	Visual
107.	Burr	Visual
108.	Foreign matter	Visual
MINOR		
	None defined	

4.2.2.4 Sleeve, Adjusting, Drawing 2218-23 covering a detail of Drawing 2218-18

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Pitch diameter of thread, max.	Gage
102.	Minor diameter of thread, max.	Gage
103.	Metal defective	Visual
104.	Protective finish inadequate or defective	Visual
105.	Lacquer stripe missing	Visual
106.	Surface finish improper	Visual
107.	Burr	Visual
108.	Foreign matter	Visual
MINOR		
	None defined	

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4.2.2.5 Tube, Intermediate, Drawing 2218-29 covering a detail of Drawing 2218-18

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Pitch diameter of threads, max., 2 holes	Gage
102.	Minor diameter of thread, max.	Gage
103.	Metal defective	Visual
104.	Protective finish inadequate or defective	Visual
105.	Solid film lubricant inadequate or defective	Visual
106.	Surface finish improper	Visual
107.	Burr	Visual
108.	Foreign matter	Visual
MINOR		
	None defined	

4.2.2.6 Tube, Anti-Rotation, Drawing 2218-30 covering a detail of Drawing 2218-18

CRITICAL		
	None defined	
MAJOR		
101.	Pitch diameter of threads, max.	Gage
102.	Minor diameter of thread, max.	Gage
103.	Splines damaged	Visual
104.	Metal defective	Visual
105.	Protective finish inadequate or defective	Visual
106.	Solid film lubricant inadequate or defective	Visual
107.	Surface finish improper	Visual
108.	Burr	Visual
109.	Foreign matter	Visual
MINOR		
	None defined	

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4.2.2.7 Tube, Ballnut Adapter, Drawing 2218-31 covering a detail of Drawing 2218-18

Defect		Method of Inspection
CRITICAL		
None defined		
MAJOR		
101.	Metal defective	Visual
102.	Protective finish inadequate or defective	Visual
103.	Solid film lubrication inadequate or defective	Visual
104.	Surface finish improper	Visual
105.	Burr	Visual
106.	Foreign matter	Visual
MINOR		
None defined		

4.2.2.8 Tube, Outside, Drawing 2218-34 covering a detail of Drawing 2218-18

CRITICAL		
None defined		
MAJOR		
101.	Pitch diameter of large outside thread, min.	Gage
102.	Major diameter of large outside thread, min.	Gage
103.	Pitch diameter of small outside thread, min.	Gage
104.	Major diameter of small outside thread, min.	Gage
105.	Splines damaged	Visual
106.	Metal defective	Visual
107.	Protective finish improper	Visual
108.	Surface finish improper	Visual
109.	Burr	Visual
110.	Foreign matter	Visual
MINOR		
None defined		

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4.2.2.9 End Cover, Drawing 2218-38 covering a detail of Drawing 2218-18

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Surface finish improper	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
None defined	

4.2.2.10 Sleeve, Holder, Drawing 2218-46 covering a detail of Drawing 2218-18

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread, max.	Gage
102. Minor diameter of thread, max.	Gage
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Surface finish improper	Visual
106. Burr	Visual
107. Foreign matter	Visual
MINOR	
None defined	

4.2.2.11 Cover, Housing, Drawing 2218-54 covering a detail of Drawing 2218-18

CRITICAL	
None defined	

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4.2.2.11 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Surface finish improper	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
None defined	

4.2.2.12 Gear, Housing, Drawing 2218-75 covering a detail of Drawing 2218-18

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of large inside thread, max.	Gage
102. Minor diameter of large inside thread, max.	Gage
103. Pitch diameter of outside thread on boss, min.	Gage
104. Major diameter of outside thread on boss, min.	Gage
105. Metal defective	Visual
106. Protective finish inadequate or defective	Visual
107. Surface finish improper	Visual
108. Burr	Visual
109. Foreign matter	Visual
MINOR	
None defined	

4.2.2.13 Head, Firing, Drawing 2218-87 covering a detail of Drawing 2218-18

CRITICAL	
None defined	

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4.2.2.13 (Cont'd)

Defect		Method of Inspection
MAJOR		
101.	Pitch diameter of large inside thread, max.	Gage
102.	Minor diameter of large inside thread, max.	Gage
103.	Pitch diameter of small inside thread, max.	Gage
104.	Minor diameter of small inside thread, max.	Gage
105.	Metal defective	Visual
106.	Protective finish inadequate or defective	Visual
107.	Surface finish improper	Visual
108.	Burr	Visual
109.	Foreign matter	Visual
MINOR		
None defined		

4.2.2.14 Retainer, Drawing 2218-108 covering a detail of Drawing 2218-18

CRITICAL		
None defined		
MAJOR		
101.	Any part missing	Visual
102.	Improper assembly	Visual
103.	Metal defective	Visual
104.	Protective finish inadequate or defective	Visual
105.	Surface finish improper	Visual
106.	Burr	Visual
107.	Foreign matter	Visual
MINOR		
None defined		

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4.2.2.15 Switch, Housing Assembly, Drawing 2218-109 covering a detail of Drawing 2218-18

Defect		Method of Inspection
CRITICAL		
None defined		
MAJOR		
101.	Any part missing	Visual
102.	Improper assembly	Visual
103.	Metal defective	Visual
104.	Protective finish inadequate or defective	Visual
105.	Surface finish improper	Visual
106.	Burr	Visual
107.	Foreign matter	Visual
MINOR		
None defined		

4.2.2.16 Cartridge Assembly, Drawing 1146-221 covering a detail of Drawing 2218-18

CRITICAL		
1.	Mixed cartridges	Visual
2.	Metal defective	Visual
3.	Sealing disc missing or damaged	Visual
4.	Marking missing, incorrect or illegible	Visual
MAJOR		
101.	Foreign matter	Visual
MINOR		
201.	Case scratched, wrinkled, bulged or dented	Visual
202.	Protective finish missing or inadequate	Visual

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4.2.2.17 Assembly, Drawing 2218-18

Defect	Method of Inspection
CRITICAL	
1. Incorrect clearance under clevis	Gage
2. Improper assembly	Visual
3. Any part missing	Visual
4. Labels missing, incorrect or illegible	Visual
MAJOR	
101. Improper alignment	Visual
102. Set screw missing, loose, above flush, or not covered with sealing compound	Visual
MINOR	
201. Location of labels incorrect	Visual
202. Protective finish missing or inadequate	Visual
203. Shipping cap missing	Visual

4.2.2.18 Carton, Packing Ammunition, Drawing C9233331

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Flaps or cover loose or distorted	Manual and Visual
202. Contents loose	Manual
203. Sealing strips missing, loose or improperly applied	Visual
204. Cut or damaged through all impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual

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4.2.2.19 Box, Packing, Ammunition, Drawing C9233334

Defect		Method of Inspection
CRITICAL		
None defined		
MAJOR		
None defined		
MINOR		
201.	Contents loose	Manual
202.	Hardware missing, loose, broken, or improperly engated	Manual and Visual
203.	Strapping missing, loose, broken or improperly assembled	Manual and Visual
204.	Contents of box exposed	Visual
205.	Marking missing, incorrect or illegible	Visual
206.	Number of units incorrect	Visual
207.	Car seal missing, unsealed or improperly positioned	Visual

4.2.3 Inspection.- The following tests shall be performed in accordance with 4.3.

4.2.3.1 Assembly torque.- Torque settings of each assembly as shown in Figure 2-7, T.O. 11P4-11-3 shall be subjected to the torque application. Failure of any assembly to comply with the requirement of 3.2 shall be cause for rejection of the assembly.

4.2.3.2 Protective finish, replated components.- The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable Military Standard listed on the drawing or T.O. 11P4-11-3.

4.2.3.3 Functional tests.

4.2.3.3.1 Motor.- Each motor shall be subjected to the test specified in 4.3.3. Failure of motor to comply with the requirements of 3.4.1 shall be cause for rejection of the motor.

4.2.3.3.2 Air firing.- One remover of each twenty or less shall be subjected to the air firing test specified in 4.3.4. Failure of remover to comply with the requirements of 3.4.4 shall be cause for rejection of the lot.

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4.2.3.3.3 Operating load.- Each remover shall be cycled electrically as specified in 4.3.5 to the number of cycles required by T.O. 11P4-11-3 to determine compliance with the requirements specified in 3.4.2.1 and T.O. 11P4-11-3. Failure of any remover to comply with the requirements of 3.4.2, 3.4.2.1 and T.O. 11P4-11-3 shall be cause for rejection of the remover.

4.2.3.3.4 Bearing center dimensions.- Each remover shall be subjected to the test for center dimensions during the cycling test specified in 4.3.5. Failure of any remover to comply with the requirements of 3.4.3 and T.O. 11P4-11-3 shall be cause for rejection.

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed in T.O. 11P4-11-3 and as specified herein.

4.3 Test methods and procedures.

4.3.1 Assembly torque.- A torque wrench, set at the applicable torque requirements specified in Figure 2-7, T.O. 11P4-11-3 shall be used in conjunction with a special adapter to determine the adequacy of the settings. The remover shall be secured in a holding device and the assembly torque applied.

4.3.2 Protective finish, replated components.- Test procedures for protective finish shall be in accordance with the applicable Military Standards referenced on the drawing or T.O. 11P4-11-3.

4.3.3 Motor.- The motor shall be installed in the test fixture specified in paragraph 3.3, T.O. 11P4-11-3. The motor shall be tested in accordance with the functional requirements of 3.4.1 and T.O. 11P4-11-3.

4.3.4 Air firing test.- Utilize the air firing test fixture specified in T.O. 11P4-11-3. An air pressure hose shall be connected to the fixture to actuate the remover under test. A pressure sensitive device shall be used to measure the pressure exerted to actuate the remover under test. The pressure sensitive device shall be inserted into the system by means of a Tee coupling between the remover and the hose connection. The remover shall be tested in accordance with the functional requirements of 3.4.4 and T.O. 11P4-11-3 and the results recorded.

4.3.5 Operating load test.- The remover shall be installed in test fixture 10553501 or in the test fixture specified in paragraph 3.2, T.O. 11P4-11-3 with special adapters associated control panels for this test. The fixture shall completely cycle the remover in accordance with the functional requirements of 3.4.2 and 3.4.2.1 with the time for linear extension and motor current drain recorded. Dimensional measuring equipment shall be used to measure the bearing center dimensions during each cycle and the results recorded.

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5. PREPARATION FOR DELIVERY

5.1 Packing and marking (Level A).- Packing and marking shall be in accordance with drawings C9233331 and C9233334.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Remanufactured exterior components are listed in Table IV.

TABLE IV

Name of Part	Part Number
Clevis, Mounting	2218-22
Sleeve, Adjusting	2218-23
Tube, Intermediate	2218-29
Tube, Anti-Rotation	2218-30
Tube, Ballnut Adapter	2218-31
Tube, Outside	2218-34
End, Cover	2218-38
Sleeve, Holder	2218-46
Cover, Housing	2218-54
Gear, Housing *	2218-75
Head, Firing	2218-87
Retainer	2218-108
Switch Housing *	2218-109

* These components shall only be replated if obviously necessary.

Custodian:

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Preparing activity:

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